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RESEARCH ARTICLE

Interannual rainfall variability promotes species coexistence primarily through relative nonlinearity in interaction strength

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Abstract

1. Combating the effects of global change and biodiversity loss requires a deeper understanding of species coexistence across variable environments. Little is known, however, about how drivers of coexistence respond to environmental change.
2. Using a greenhouse competition experiment, we examine how interannual rainfall variation influences competitive dynamics in an old-field annual plant community in southern China. We parameterize a competition model for eight species to assess coexistence outcomes under constant and variable rainfall conditions, partitioning variation-independent and variation-dependent mechanisms of coexistence.
3. We show that species are more likely to coexist under variable rainfall conditions. Of the coexistence mechanisms examined, relative nonlinearity of the interaction strength most strongly promotes coexistence, followed by a lesser positive contribution from relative nonlinearity of the vital rates, while the interaction effect between these nonlinearities tends to destabilize coexistence.
4. *Synthesis*. Our study provides novel evidence that rainfall variability promotes pairwise species coexistence in a multispecies plant community, driven primarily by the relative nonlinearity in interaction strength. These findings highlight the necessity of incorporating environmental variability into coexistence studies to better predict and manage biodiversity in a changing world.

KEYWORDS
biodiversity maintenance, environmental variability, modern coexistence theory, plant–plant interactions, relative nonlinearity, storage effect, temporal fluctuation

1 | INTRODUCTION

Understanding the mechanisms underpinning diversity maintenance is a long-standing focus of ecology and conservation biology. Global change and biodiversity loss make it increasingly important to fill this knowledge gap (Cardinale et al., 2012; Naeem et al., 1994; Rockström et al., 2009). The key question of how species coexist to form diverse communities remains poorly

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understood (Chesson, 2000; Gause, 1934; Hutchinson, 1961). Following modern coexistence theory, much empirical research has predicted the long-term outcomes of biotic interactions based on niche differences (reflecting resource partitioning) and fitness differences (reflecting competitive asymmetry) between species (Adler et al., 2007; Buche et al., 2022; Godoy & Levine, 2014; Levine & HilleRisLambers, 2009; Narwani et al., 2013). When niche differences exceed fitness differences, species can theoretically coexist stably (Chesson, 1990). However, these predictions often fail to predict the local species richness observed in nature, even among commonly co-occurring species (Kraft et al., 2015; Matías et al., 2018; Wainwright et al., 2019). One key reason is likely that coexistence models in these studies commonly assume species interactions occur in constant environments, and therefore, only make accurate predictions if the environment remains unchanged over time (Chesson, 2000; Hallett et al., 2019; Hart et al., 2017). Though environmental variation is ubiquitous in nature (Chesson, 2000; Chesson & Warner, 1981; Gravel et al., 2011; Hutchinson, 1961; Letten et al., 2018; Usinowicz et al., 2017), and becoming more extreme under climate change, understanding how it impacts community diversity through coexistence dynamics remains an underexplored but critical question.

Understanding how environmental variation impacts coexistence requires an approach to distinguish its effects on different components of the coexistence process from variation-independent mechanisms. A parallel framework of modern coexistence theory achieves this by decomposing invasion growth rates—the per-capita growth rates when rare (Barabás et al., 2018; Chesson, 1994, 2000). Two species may coexist as long as each can invade a monoculture equilibrium of the other (e.g. positive invasion growth rates) (Bowler et al., 2022; Chesson, 2000, 2018; Chesson & Warner, 1981; Macarthur & Levins, 1967). By decomposing mean long-term invasion growth rates in varying environments, we can distinguish environmental variation-independent from variation-dependent mechanisms, notably relative nonlinearity and the storage effect (Barabás et al., 2018; Chesson, 2000; Ellner et al., 2016; Letten et al., 2018).

Relative nonlinearity occurs when species exhibit differential nonlinear responses in their population growth rates to environmental (density-independent) and competitive (density-dependent) variation (Armstrong & McGehee, 1980; Chesson & Rosenzweig, 1991; Letten et al., 2018; Levins, 1979; Tredennick et al., 2017). Nonlinear responses to environmental variation encompass (1) direct demographic responses (i.e. relative nonlinearity in vital rates) and (2) indirect effects via biotic interactions (i.e. relative nonlinearity in interaction coefficients; Chesson, 2020; Ellner et al., 2019; Werner et al., 2025). Specifically, species with more convex response curves (i.e. accelerating growth with resources) benefit from variability, whereas those with less convex responses are disadvantaged due to nonlinear averaging (Chesson, 1994; Levins, 1979). These asymmetries can promote coexistence by reducing fitness inequalities between species (Chesson, 2000; Letten et al., 2018; Yuan & Chesson, 2015). Besides, nonlinear responses to competitive variation can promote coexistence when the species with more convex responses dampen variation while the species with less convex responses amplify it (Armstrong & McGehee, 1980; Barabás et al., 2018; Chesson, 2000).

The storage effect arises from the interaction between variation in environment and variation in competition (Chesson, 2000; Chesson & Huntly, 1989). Coexistence via the storage effect depends on three key ingredients: (1) species-specific environmental responses, (2) sub-additive effects of competition with the environment (e.g. species are more sensitive to competition during favourable conditions) and (3) positive covariance between environment and competition, driven by strong causal links between environment and competition and a temporally autocorrelated environment (Chesson, 1994; Chesson & Huntly, 1989; Johnson & Hastings, 2022; Stump & Vasseur, 2023). The storage effect has been understood as representing temporal resource partitioning and potentially positively contributing to species coexistence, thus it has received more attention in the literature (Angert et al., 2009; Chesson, 1983; Johnson & Hastings, 2022; Usinowicz et al., 2017). In contrast, relative nonlinearity, despite its potential significance, has been largely overlooked in empirical research (Letten et al., 2018; Stump & Vasseur, 2023; Zepeda & Martorell, 2021).

Despite the widely recognized importance of environmental variability for species coexistence and the well-established theoretical framework for quantifying variation-dependent coexistence mechanisms, empirical studies remain scarce (Hallett et al., 2019; Muehleisen et al., 2024; Shoemaker et al., 2020). Previous studies either relied on field data, which are often confounded by multiple interacting factors or have been limited to a handful of species. A major challenge is that quantifying these mechanisms requires estimating interaction coefficients across multiple environmental conditions, a task that becomes increasingly demanding as the number of species increases (Hart et al., 2018; Inouye, 2001). Another challenge is the gap between modern coexistence theory and empirical applications, as the theory relies on restrictive assumptions and requires complex mathematical derivations to adapt to specific study systems (Barabás et al., 2018). Recent advances in simulation-based methods have significantly simplified analyses of such datasets, making it easier to decompose invasion growth rates and quantify different coexistence mechanisms (Ellner et al., 2016, 2022).

Variability in rainfall is increasing in many regions with climate change, making it a likely driver of future community-level changes through variation-dependent coexistence mechanisms (Allen & Ingram, 2002; Donat et al., 2016; Li et al., 2025). Studies on coexistence under different rainfall regimes have shown that coexistence outcomes depend on water availability and that this effect works variably through interactions and intrinsic fitness (Matías et al., 2018; Van Dyke et al., 2022; Wainwright et al., 2019). However, these studies primarily compared outcomes under constant rainfall scenarios—ambient versus sustained droughts—rather than incorporating interannual variability. This is

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particularly important to understand in the face of climate change. Without considering the historical influence of interannual rainfall variability on coexistence, we cannot accurately predict how future changes to rainfall regimes will alter the ability of species to coexist.

In order to unravel how rainfall variability shapes coexistence, we set up a greenhouse experiment to parameterize mutual invasibility-based coexistence models for eight annual herbaceous plant species under four distinct annual rainfall treatments: consistent dry, consistent wet, post-flood dry and pre-flood dry. By simulating two-species population dynamics over multiple years, we aimed to determine how coexistence outcomes under constant interannual rainfall conditions differ from rainfall regimes that vary interannually. Our study addresses the following questions: (1) What is the impact of interannual rainfall variability on coexistence? (2) Which mechanisms of coexistence are most important under variable compared to constant rainfall conditions?

2 | METHODS

2.1 | Study site and species selection

The greenhouse experiment was conducted at the He'er Kou Town, Fengkai County, Zhaoqing City, Guangdong Province, China (23°30′57.2″ N, 111°47′49.7″ E). The site experiences a subtropical monsoon climate characterized by hot and humid summers, as well as cool and dry winters. Approximately 78% of annual rainfall occurs between April and October, aligning with the growing season. Within this timeframe, April to June constitutes the pre-flood season, characterized by monsoon-influenced rainfall, while July to September marks the post-flood season, primarily influenced by typhoons (Figure S1a). The amount of rainfall during these two seasons varies from year to year. On average, the pre-flood season receives 243.9 mm of rainfall per month (ranging from 39.2 to 559 mm), while the post-flood season receives 178.3 mm of rainfall (ranging from 15.8 to 517.3 mm).

We extracted 120-year monthly rainfall data (1901–2020) for He'er Kou Town from a 1 km dataset provided by the National Earth System Science Data Center, National Science & Technology Infrastructure of China (http://www.geodata.cn/). These data were classified into four rainfall types: consistent dry (below-average rainfall across seasons), consistent wet (above-average rainfall across seasons), post-flood dry (above-average rainfall in pre-flood season, below-average rainfall in post-flood season) and pre-flood dry (below-average rainfall in pre-flood, above-average rainfall in post-flood season) (Figure S1b). The rainfall time series showed no significant temporal autocorrelation at short lags (Figure S2; Ljung-Box test, $p = 0.491$).

The representative vegetation in the area is natural old-field vegetation. For our experiment, we selected eight common annual herbaceous plant species that naturally co-occur at the study site (Table S1). These include four native species (*Dactyloctenium aegyptium*, *Oplismenus compositus*, *Ludwigia hyssopifolia* and *Polygonum hydropiper*) and four invasive species (*Ageratum conyzoides*, *Bidens pilosa*, *Physalis angulata* and *Praxelis clematidea*). No specific permits were required for this study.

2.2 | Experimental design

We collected seeds at the He'er Kou Town in autumn 2021 and sowed them in seedling trays with sterile vermiculite in April 2022. In May 2022, seedlings were transplanted into pots [12 × 12 × 9.7 cm] filled with soil sourced from the area surrounding the greenhouse. The soil was sieved through a 10-mesh sieve and air-dried before being used for the experiment. A response surface experimental design was adopted for all 28 species pairs from 8 species (Inouye, 2001; Law & Watkinson, 1987), with each species grown in monoculture and together in a 1:1 mixture at three density levels (2, 4, 8 plants/pot). These density levels correspond to roughly 139, 278 and 556 plants/m^2 (the density around the study site was about 1400 plants/m^2). Each treatment comprised 108 pots, with two replicates and four treatments, totaling 864 pots, 4032 individuals. Dead seedlings were replaced within 1 week after seedling transplantation; then, we implemented four rainfall treatments.

From 120-year rainfall data, we calculated mean precipitation for entire pre- and post-flood periods in both wet and dry scenarios—833 mm (wet pre-flood), 624 mm (dry pre-flood), 643 mm (wet post-flood) and 441 mm (dry post-flood)—and adjusted watering based on pot area. Specifically, we applied 270 mL every 2 days for wet pre-flood and 200 mL for dry pre-flood periods, and 200 mL for wet post-flood and 150 mL for dry post-flood periods. On non-rainy days, the greenhouse was opened for ventilation and pollination.

We assessed fecundity by measuring seed yield of all plants in each pot. Given variations in phenology, seeds of *D. aegyptium* and *P. angulata* were collected in late August, *Po. hydropiper* and *Pr. clematidea* in late September, *B. pilosa* and *A. conyzoides* in mid-October and *O. compositus* and *L. hyssopifolia* in late November. Fecundity was calculated as the product of fruit or infructescence counts and their average seed number.

To evaluate seed germination fraction (g) and seed bank survival (s), mature seeds were collected from abandoned farmland in the He'er Kou Town in autumn 2021. In April 2022, we conducted a greenhouse germination experiment under simulated wet and dry pre-flood conditions. For each species, 2000 seeds were sown in seedling trays [54 × 28 × 5 cm] filled with sterilized vermiculite, and watered daily with 1400 mL (wet) or 1000 mL (dry). In May 2022, seedling emergence was recorded, and germination fraction (g) was calculated as the proportion of emerged seedlings relative to sown seeds. For seed bank survival (s), three nylon mesh bags containing seeds of each species were buried in April 2022 and retrieved in January 2023. Viability of 100 seeds per species was tested before and after burial using germination trials and triphenyl tetrazolium chloride (TTC) staining, with three replicates. Survival was calculated as the ratio of viable seeds post-burial to pre-burial.

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2.3 | Model parameterization

To estimate population growth rates of species under different rainfall scenarios, we parameterized a discrete-time model describing the dynamics of annual plant populations (Ellner, 1984; Levine & Rees, 2004; Pacala, 1986; Watkinson, 1980). The population growth rate of focal species *i* is described by the following equation:

$$\frac{N_{i,t+1}}{N_{i,t}} = (1 - g_{i,t})s_i + g_{i,t}F_{i,t} \quad (1)$$

where $N_{i,t}$ is the number of viable seeds of species *i* per pot before germination at time *t*. The first term on the right-hand side represents the carryover of seeds in the seed bank, where s_i is the annual survival of seeds in the seed bank and $g_{i,t}$ is the fraction of seeds that germinate at time *t*. The second term represents annual seed production, where $F_{i,t}$ is the number of viable seeds produced per germinated individual for species *i* at time *t*. $F_{i,t}$ can be expressed as a function that describes how fecundity responds to intraspecific and interspecific interactions. We used the Ricker model due to its widespread use in similar systems and its compatibility with positive interspecific interactions (Bimler et al., 2018; Inouye, 2001; Ricker, 1954; Zhang & van Kleunen, 2019). We conducted a formal model comparison against three alternatives, Beverton-Holt, Lotka-Volterra and Law-Watkinson, using the Watanabe-Akaike Information Criterion (WAIC) to confirm that the Ricker model was also an appropriate statistical choice, as it provided the best fit for all eight species (see Appendix S2, Table S5). The model is expressed as:

$$F_{i,t} = \lambda_{i,t} e^{-a_{ii,t}g_{i,t}N_{i,t} - a_{ij,t}g_{j,t}N_{j,t}} \quad (2)$$

where $\lambda_{i,t}$ represents the seed production in the absence of neighbours at time *t*, while $a_{ii,t}$ and $a_{ij,t}$ describe the per-capita intraspecific and interspecific interaction effect on fecundity, respectively. We employed a normal likelihood function to model fecundity, paired with weakly informative priors ($\lambda \sim$ Normal (100, 10,000); $a \sim$ Normal (0.1, 5)). The model was fitted for each of the four rainfall treatments using the *ulam* function from the *rethinking* package in R version 4.0, with 4 chains, 4000 iterations (1000 warmup). The parameters λ and intraspecific interaction a_{ii} were constrained to be positive to ensure positive seed production and prevent self-facilitation, respectively. The posterior distributions of the estimated parameters λ and a are provided in Figure S3. Posterior predictive checks indicate that the predicted values of seed production generally align well with observations (Appendix S2, Figures S7 and S8). Yet, the model underestimates the frequency of non-reproducing individuals, which may potentially lead to an overestimation of invasion growth rates and coexistence probabilities.

Pots were included as a random effect to account for the non-independence of individuals within the same pot (Zhang & van Kleunen, 2019). However, incorporating the random effect led to elevated $\hat{R}$(statistics used to assess the convergence of Markov Chain Monte Carlo simulations) for some parameters, indicating potential convergence issues. Specifically, the median $\hat{R}$ was 1.00, suggesting good convergence for most parameters, but 12 out of 288 parameters (4.17%) had $\hat{R}$ values exceeding 1.06, with the highest reaching 1.44. In contrast, the model without random effects showed good convergence, with a median $\hat{R}$ of 1.00 and a maximum value of 1.02. For the model with random effects, traceplot inspection of parameters with $\hat{R}$ values exceeding 1.06 revealed that most chains remained closely overlapping, despite bimodality in a few cases. Comparisons of parameter estimates between models with and without the random effect revealed no significant differences (see Appendix S3, Figure S9). Furthermore, results from the model with the random effect exhibited consistent trends with those without (Appendix S3, Figures S10–S13), supporting the robustness of our conclusions. Therefore, we report the results from the model without the random effect in the main text.

2.4 | Invasion growth rate analysis for coexistence under four manipulated rainfall scenarios

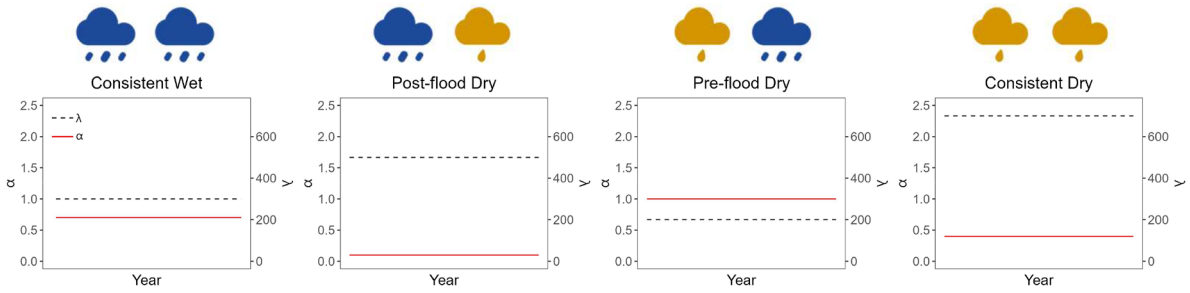
To evaluate species coexistence outcomes under different rainfall regimes, we estimated the mutual invasion growth rates for each species pair under four constant rain scenarios, respectively (Figure 1a). Note that this approach simplifies interannual rainfall variation by representing it with four discrete regimes. Stable coexistence requires that the invasion growth rates of both species are positive; otherwise, species cannot coexist. To calculate the invasion growth rate, we first simulated the population dynamics of the resident species over a 120-year period under each constant rainfall scenario, using the equilibrium population sizes as the initial conditions. The first 60 years were treated as a burn-in phase to allow the resident's population growth rates to stabilize, after which the invader was introduced. The invasion growth rate was computed as the average growth rate over the remaining 60 years (Figure S4).

FIGURE 1 Conceptual illustration of invasion simulation scenarios and evaluation of the contribution of different coexistence mechanisms to invasion growth rate. Based on four sets of population parameters (α_1, λ_1; α_2, λ_2; α_3, λ_3; α_4, λ_4) obtained from competition experiments under four manipulated rainfall treatments, (a) we simulated population dynamics under four constant rainfall scenarios to calculate mutual invasion growth rates and evaluate coexistence outcomes in each scenario. Black dashed lines represent λ, and red solid lines represent α. (b) To decompose the invasion growth rates, we constructed four additional scenarios: Historical rainfall fluctuations, both α and λ constant, variable α with constant λ and constant α with variable λ. Here, $\overline{\alpha}$ and $\overline{\lambda}$ are the weighted averages of the parameters under the four rainfall treatments. (c) We decomposed the population growth rates into four components ϵ^0, ϵ^α ϵ^λ and $\epsilon^{\alpha\lambda}$. (d) Finally, we performed term-by-term invader-resident comparison and partitioned the invasion growth rates under environmental fluctuations into four coexistence mechanisms: Δ_i^0, representing the variation-independent mechanism, Δ_i^α, the relative nonlinearity in interaction coefficients; Δ_i^λ, the relative nonlinearity in vital rates; and $\Delta_i^{\alpha\lambda}$, quantifying the interaction effect between variability in α and λ.

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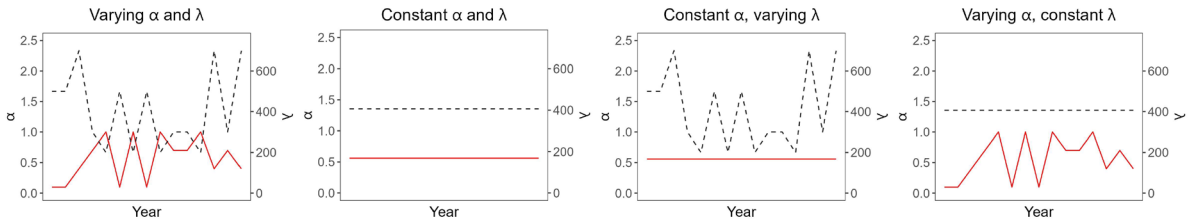
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(a) Invasion analysis for coexistence outcomes under four constant rainfall scenarios



$r(\alpha_1, \lambda_1)$ $r(\alpha_2, \lambda_2)$ $r(\alpha_3, \lambda_3)$ $r(\alpha_4, \lambda_4)$

(b) Invasion analysis for coexistence mechanisms partitioning



$r(\alpha, \lambda)$ $r(\bar{\alpha}, \bar{\lambda})$ $r(\bar{\alpha}, \lambda)$ $r(\alpha, \bar{\lambda})$

(c) Population growth rate partitioning

$$r(\alpha, \lambda) = \epsilon^0 + \epsilon^\alpha + \epsilon^\lambda + \epsilon^{\alpha\lambda}$$

$\epsilon^0 = r(\bar{\alpha}, \bar{\lambda})$ ..... Null growth rate with no variation

$\epsilon^\alpha = r(\alpha, \bar{\lambda}) - r(\bar{\alpha}, \bar{\lambda})$ ..... Main effect of variation in α

$\epsilon^\lambda = r(\bar{\alpha}, \lambda) - r(\bar{\alpha}, \bar{\lambda})$ ..... Main effect of variation in λ

$\epsilon^{\alpha\lambda} = r(\alpha, \lambda) - (\epsilon^0 + \epsilon^\alpha + \epsilon^\lambda)$ ..... Interaction between variation in both α and λ

(d) Quantifying coexistence mechanism through invader-resident comparisons

$$\bar{r}_i \approx \bar{r}_i - \bar{r}_r = (\epsilon_i^0 - \epsilon_r^0) + (\epsilon_i^\alpha - \epsilon_r^\alpha) + (\epsilon_i^\lambda - \epsilon_r^\lambda) + (\epsilon_i^{\alpha\lambda} - \epsilon_r^{\alpha\lambda})$$

$$= \Delta^0 + \Delta^\alpha + \Delta^\lambda + \Delta^{\alpha\lambda}$$

Variation independent mechanism

Relative nonlinearity in α

Relative nonlinearity in λ

Interaction effect

Variation dependent mechanisms

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To account for parameter uncertainty, we sampled 1000 parameter sets from the posterior distributions of α and λ. We then employed two complementary approaches to assess coexistence outcomes. First, we used median estimates of invasion growth rates resulting from the above sampling, with stable coexistence defined as positive median invasion growth rates for both species. Second, we calculated the coexistence probability, defined as the proportion of posterior samples in which both species had positive invasion growth rates.

Additionally, to verify the robustness of our probabilistic approach against potential uncertainty in parameters *s* and *g*, we drew 1000 samples from binomial distributions and incorporated them into simulations to calculate invasion growth rates, thereby propagating the uncertainty in *s* and *g* alongside α and λ (Appendix S4, Figures S14–S17).

2.5 | Partition of coexistence mechanisms

We applied the method developed by (Ellner et al., 2019) to decompose the contributions of different coexistence mechanisms to the invasion growth rates. To achieve this, we constructed four counterfactual scenarios: (1) historical rainfall fluctuations, where both interaction coefficients (α) and intrinsic vital rates (λ and *g*) vary according to historic rainfall, (2) variable α with constant λ and *g*, (3) constant α with variable λ and *g* and (4) constant α, λ and *g* (Figure 1b). Constant α, λ and *g* values were weighted averages of parameters from four manipulated rainfall scenarios, based on their 120-year frequency. For each scenario, we calculated the population growth rates for each species pair during mutual invasions.

Based on these population growth rates *r* from the four counterfactual scenarios, we decomposed the *r* of invader or resident under historical rainfall fluctuations into four terms and performed a term-by-term comparison (Figure 1c,d). This allowed us to partition the average invasion growth rate under historical rainfall fluctuation into the four coexistence mechanisms:

$$\overline{r_i} \approx \overline{r_i} - \overline{r_r} = \Delta_i^r = \Delta_i^0 + \Delta_i^\alpha + \Delta_i^\lambda + \Delta_i^{\alpha\lambda} \quad (3)$$

$\overline{r_i}$ represents the invasion growth rate and $\overline{r_r}$ represents the growth rate of the resident, which is approximately zero. The invader-resident comparison quantified the contributions of each term to invasion growth rate by either enhancing the invader's performance or limiting the resident's.

The first partition term Δ_i^0, represents the variation-independent mechanism. Since the resident's population growth rate is near 0, Δ_i^0 can also be interpreted as the invasion growth rate in the absence of environmental variation.

The remaining three partition terms represent variation-dependent mechanisms. Δ_i^α is the relative nonlinearity in interaction coefficients, quantifying the indirect effect of environmentally driven variation on biotic interactions. While analogous to relative nonlinearity in competition from Chesson's framework—which arises from density-dependent growth responses to competitive variation—Δ_i^α differs in that it involves interaction coefficients that are not density-dependent. Δ_i^λ, the relative nonlinearity in vital rates, quantifying the direct effect of environmentally driven variation in demographic responses, including intrinsic seed production and germination fraction. Δ_i^λ is included in Chesson's MCT framework as part of the invasion growth rate in the absence of direct density-dependent feedback, we have separated it out as a distinct term in our analysis (Ellner et al., 2019). Finally, $\Delta_i^{\alpha\lambda}$ quantifies the interaction effect between variability in α and λ, accounting for their combined influence that is not explained by the main effects (Ellner et al., 2019).

3 | RESULTS

3.1 | Coexistence outcomes under constant versus variable scenarios

Variable rainfall supported the highest levels of coexistence, with an average probability of 25.1% across all species pairs and seven pairs exhibiting coexistence (defined by positive median invasion growth rates for both species, Figure 2; Figure S5). Among the five constant rainfall scenarios, the average coexistence probability was lowest under average rainfall (6.5%), corresponding to only two coexisting pairs (Figure 2; Figure S5). In contrast, the average coexistence probability was highest under post-flood dry (24.1%), with five coexisting pairs (Figure 2; Figure S5).

We found that how rainfall variation influenced coexistence outcomes varied depending on the identity of species in species pairs. Specifically, coexistence probability was higher under variable rainfall than under constant average rainfall for 17 out of 28 species pairs (Figure 2e,f; Table S2). The remaining pairs either showed negligible differences or zero coexistence probability under both regimes (Figure 2e,f; Table S2).

Across all pairs, the competitive outcome had a 26.1% probability of changing when shifting from constant average to variable rainfall conditions. Among these outcome transitions, 74.1% were from non-coexistence to coexistence, whereas only 2.8% resulted in a loss of coexistence, and the remaining involved transitions between non-coexistence states (Figure 2e,f). Based on median estimates, rainfall variability triggered competitive outcome transitions in seven pairs: Five involved shifts across the boundary from competitive exclusion to coexistence, while the remaining two involved transitions between different non-coexistence states (Table S2). For *Po. hydropiper*/*Pr. clematidea*, median-based outcomes revealed a reversal in competitive dominance—from *Po. hydropiper* excluding *Pr. clematidea* under constant conditions (82.7%) to the reverse under variability (60.2%). For *B. pilosa*/*O. compositus*, variability shifted the most probable outcome from priority effect (56.1%) to *B. pilosa* dominance (78%) (Figure 2e,f).

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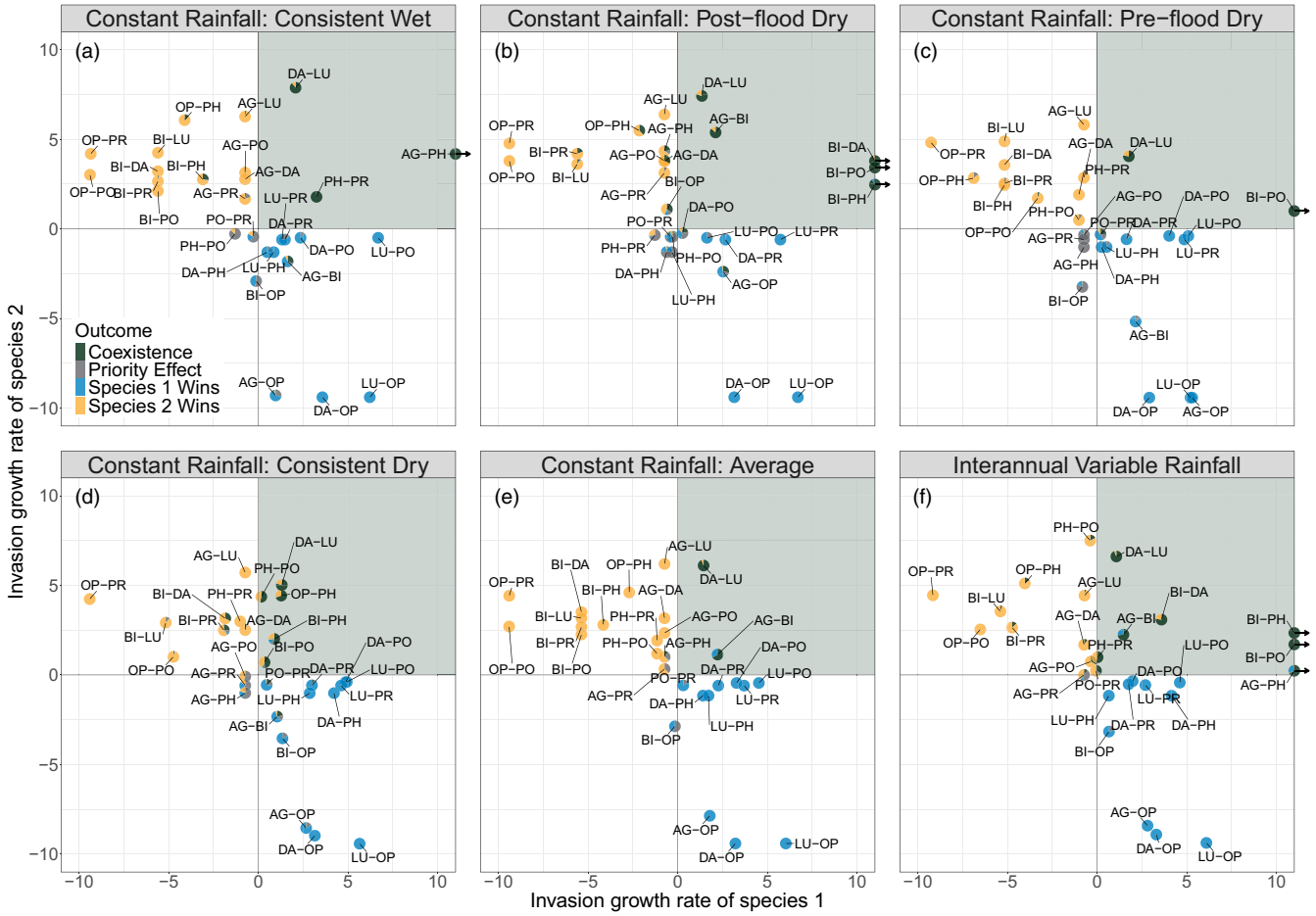


FIGURE 2 Pairwise coexistence outcomes under constant and variable rainfall scenarios. Panels (a–e) represent interannually constant rainfall scenarios, including (a) consistent wet, (b) post-flood dry, (c) pre-flood dry, (d) consistent dry and (e) average (mean across the four scenarios above), while panel (f) displays the variable interannual rainfall scenario. Points denote median invasion growth rates of species 1 and 2 (first and second in pair labels) when they invade each other's equilibrium population, calculated from 1000 parameter combinations drawn from posterior distributions. Pie charts overlaid on each point depict the posterior probabilities of each coexistence outcome (coexistence, priority effect, species 1 wins and species 2 wins). Pairs coexist when both invasion growth rates exceed 0, as indicated by the light green region. Pairs with invasion growth rates exceeding the displayed range are marked with arrows. Species codes are AG (*Ageratum conyzoides*), BI (*Bidens pilosa*), DA (*Dactyloctenium aegyptium*), LU (*Ludwigia hyssopifolia*), OP (*Oplismenus compositus*), PH (*Physalis angulata*), PR (*Praxelis clematidea*), PO (*Polygonum hydropiper*).

3.2 | The contribution of different variation-dependent mechanisms to coexistence

Although variation-dependent mechanisms generally enable more species pairs to coexist, different mechanisms may contribute to this outcome in distinct ways (Figure 3). To evaluate their relative importance, we partitioned the variation-dependent mechanisms and quantified the contributions of relative nonlinearity in interaction coefficients (Δ^{α}), relative nonlinearity in vital rates (Δ^{λ}) and their interaction ($\Delta^{\alpha\lambda}$) to coexistence probability.

Relative nonlinearity in interaction coefficients (Δ^{α}) was the dominant driver of coexistence in this study. Across all pairs, Δ^{α} increased the invasion growth rates of both species with a probability of 29.1% and inhibited both with 5.9% (Figure 3a). Notably, in the remaining 65% where Δ^{α} had opposing effects on each species, it favoured the weaker competitor with a probability of 80.8%. For instance, in the absence of rainfall variability, *Ph. angulata* could not invade *Pr. clematidea*'s stable population (invasion probability = 0%), whereas *Pr. clematidea* could invade *Ph. angulata* (100%). Δ^{α} increased *Ph. angulata*'s invasion probability to 54.2% while marginally reducing *Pr. clematidea*'s to 99.5% (Figure S6). Across paired posterior draws, this yielded a 53.8% probability that the pair switched from competitive exclusion to coexistence (Table S3). Without Δ^{α}, four of the five pairs that transitioned to coexistence under historical rainfall variation would no longer coexist (median invasion growth rates for both species <0) (Table S4).

In contrast, relative nonlinearity in vital rates (Δ^{λ}) made a more limited overall contribution. It simultaneously decreased invasion growth rates in 25.1% of cases, while increasing both in only 13.6% (Figure 3b). In the 43.4% where Δ^{λ} had opposing effects on the two species, it favoured the weaker competitor with a probability of 70.7%. Despite its relatively modest average effect, Δ^{λ} played an

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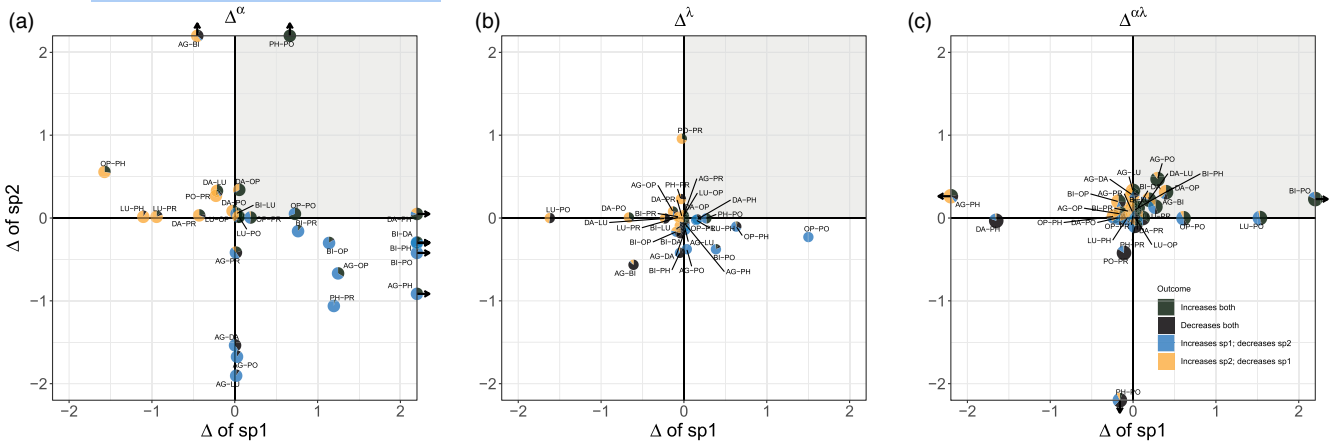


FIGURE 3 Role of different variation-dependent coexistence mechanisms. Points denote median Δ partition terms representing three variation-dependent mechanisms for species 1 and 2 (first and second in pair labels), calculated from 1000 parameter combinations drawn from posterior distributions. Pie charts overlaid on each point depict the posterior probabilities of each outcome (increased invasion growth rate for both species; decreased for both; increased for species 1 and decreased for species 2; increased for species 2 and decreased for species 1). Pairs with invasion growth rates exceeding the displayed range are marked with arrows. The variation-dependent mechanisms included (a) the main effect of environmentally driven variation in interaction coefficients (Δ^{α}, relative nonlinearity in interaction coefficients), (b) the main effect of environmentally driven variation in vital rates (Δ^{λ}, relative nonlinearity in vital rates) and (c) the interaction effect between variation in interaction coefficients and vital rates ($\Delta^{\alpha\lambda}$).

important role in particular species pairs. For example, under constant average rainfall, *Pr. clematidea* could hardly invade *Po. hydropiper* (invasion probability of 1.3%), whereas *Po. hydropiper* could invade *Pr. clematidea* (83.8%). Adding Δ^{λ} alone increased the invasion probability of *Pr. clematidea* to 99.1%, while *Po. hydropiper*'s remained nearly unchanged at 83.5%, resulting in an 81.5% posterior probability of switching from competitive exclusion to coexistence (Table S3). Nevertheless, based on median estimates, removal of Δ^{λ} did not alter the coexistence status of any species pair (Table S4).

The interaction effect ($\Delta^{\alpha\lambda}$) simultaneously increased invasion growth rates in 33.4% of cases while decreasing them in 21.1% (Figure 3c). In the remaining 45.5% where $\Delta^{\alpha\lambda}$ had opposing effects on each species, it tended to favour the stronger competitor (58.2%). While removal of $\Delta^{\alpha\lambda}$ did not change the total number of coexisting pairs based on median estimates, it did change which pairs coexisted. Specifically, its removal caused the *A. conyzoides*/*Ph. angulata* pair to transition from coexistence to *A. conyzoides* excluding *Ph. angulata* (coexistence probability falling from 66.5% to 42.3%), whereas the *Po. hydropiper*/*Pr. clematidea* pair shifted from dominance by *Pr. clematidea* to coexistence (rising from 31.8% to 56.7%) (Figure S6; Table S4).

To further assess the contribution of the three variation-dependent mechanisms to the probability of coexistence, we progressively added the mechanisms starting from a baseline that included only the variation-independent mechanism (Δ^{0}) to the full model including all mechanisms (Δ^{r}) (Figure 4). Incorporating Δ^{α} increased coexistence probability in 17 species pairs and decreased it in only one pair, raising the mean coexistence probability across all pairs from 6.5% to 24.2%. Adding Δ^{λ} then increased coexistence probability in seven pairs but decreased it in another seven, yielding a modest net positive effect on the average (+1.2%). Finally, incorporating $\Delta^{\alpha\lambda}$ increased coexistence probability for eight pairs and decreased it for seven, resulting in a marginal net reduction in average coexistence probability (−0.3%) (Figure 4).

4 | DISCUSSION

We found that interannual rainfall variation promotes species coexistence in an old-field annual plant community through distinct underlying mechanisms. The average probability of coexistence peaked under rainfall variation. Among the mechanisms examined, relative nonlinearity in interaction coefficients emerged as the key factor promoting coexistence in our study.

Environmental variability is widely considered a mechanism for biodiversity maintenance (Chesson, 2000; Levins, 1979; Snyder, 2008), yet its role in species coexistence remains hotly debated due to inconsistent evidence (Adler et al., 2006, 2009; Chu & Adler, 2015; Hallett et al., 2019; Zepeda & Martorell, 2021). While some studies suggest that climate fluctuations promote species persistence (Adler et al., 2006; Hallett et al., 2019), others report negligible effects (Adler et al., 2009; Chu & Adler, 2015; Muehleisen et al., 2024). Most prior work, however, relies on observational data or experiments with few species, leaving its general impact unclear. Our study addresses this gap through controlled pairwise competition experiments across eight annual plant species. We found that rainfall variation significantly enhanced coexistence, enabling the highest number of species pairs to coexist (Figure 2), with 7 out of 8 species coexisting with at least one competitor based on median invasion growth rate. This pattern was further reflected in coexistence probabilities, which increased in 17 out of 28 species pairs under variable rainfall compared to constant average conditions. Aside

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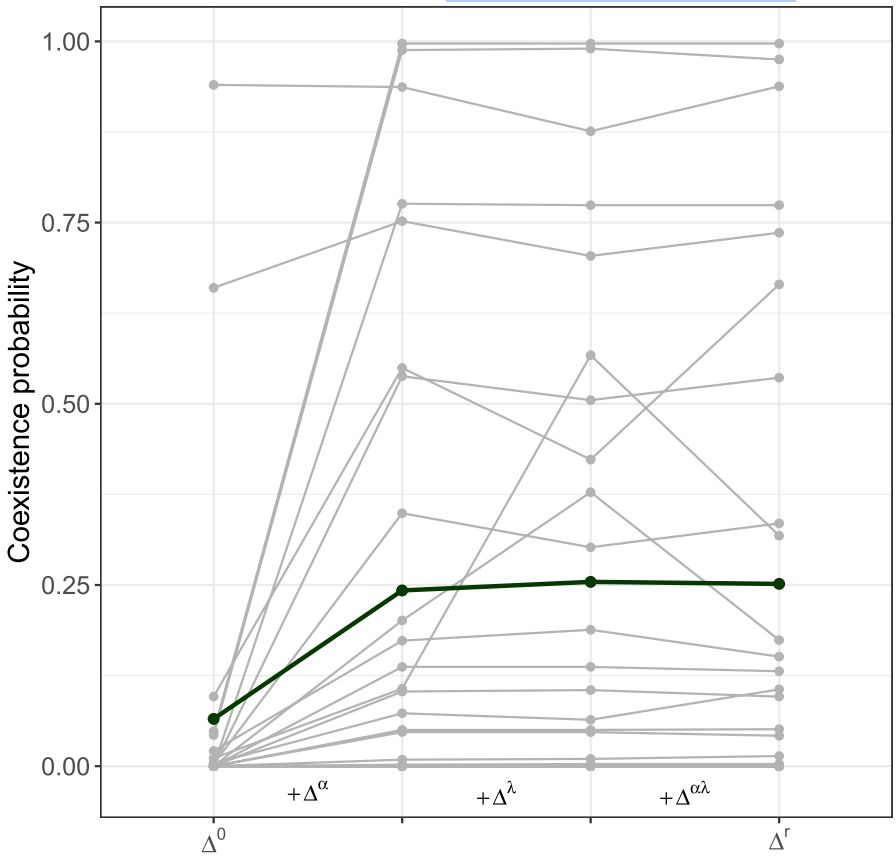


FIGURE 4 The contribution of different variation-dependent coexistence mechanisms to coexistence probability. Grey points show the coexistence probabilities for each species pair as three variation-dependent mechanisms (Δ^{α}, Δ^{λ}, $\Delta^{\alpha\lambda}$) are added stepwise from Δ^{0} to Δ^{r}. Bold green points and line represent the average coexistence probability across 28 species pairs.

from a slight decrease in *D. aegyptium*/*L. hyssopifolia*, the coexistence probability of 10 species pairs remained zero regardless of rainfall variation (Figure 4). Yet some of these pairs (e.g. *O. compositus*/*Po. hydropiper*) moved closer to the coexistence region under variable rainfall in terms of their mutual invasion growth rates (Figure 2), suggesting potential stabilization through reduced fitness asymmetries.

Empirical support for relative nonlinearity as a coexistence mechanism remains sparse and equivocal, particularly regarding its effect through density-independent environmental variation (Ellner et al., 2019; Hallett et al., 2019; Letten et al., 2018; Yuan & Chesson, 2015; Zepeda & Martorell, 2021). For example, Hallett et al. (2019) showed that relative nonlinearity in interaction coefficients stabilized coexistence for a grass-forb pair in California grasslands, while relative nonlinearity in intrinsic seed production exerted a destabilizing influence. In contrast, Muehleisen et al. (2024) found that relative nonlinearity—both in interaction coefficients and intrinsic seed production—destabilized coexistence for five annual plant species unable to coexist at the same site. Our study, spanning 28 species pairs in controlled experiments, demonstrates that relative nonlinearity in both interaction coefficients (Δ^{α}) and vital rates (Δ^{λ}, including intrinsic seed production and germination fraction) generally promotes coexistence, with the former playing a dominant role. Specifically, Δ^{α} enhanced coexistence probability in 17 out of 28 species pairs (~61%) (Figures 3 and 4). Of the remaining pairs, 10 exhibited zero coexistence probability under average rainfall, typically indicative of absolute competitive dominance by one species over the other. For instance, the native forb *L. hyssopifolia* excluded nearly all other species (except native grass *D. aegyptium*) across all scenarios. Although Δ^{λ} on average had a much weaker overall effect than Δ^{α}, it still contributed substantially in some species pairs, underscoring the need to evaluate multiple facets of relative nonlinearity to fully elucidate its influence on species coexistence.

Relative nonlinearity arising from exogenous environmental fluctuations—as opposed to endogenous population dynamics—has not traditionally been regarded as a stabilizing mechanism. Instead, it is thought to promote coexistence primarily through an equalizing effect, by elevating the invasion growth rates of inferior competitors while suppressing those of superior ones (Barabás et al., 2018; Letten et al., 2018; Yuan & Chesson, 2015). Our experiments confirm this equalizing effect of relative nonlinearity. However, in certain species pairs, relative nonlinearity also increased median invasion growth rates for both species simultaneously (Figure S6). These pairs, which exhibited the greatest increases in coexistence probability, were predominantly characterized by this pattern. Notably, the stabilizing effect was most prevalent in interactions between invasive species, less common in invasive–native species pairs and rare in native–native pairs, indicating that relative nonlinearity may facilitate invasional meltdowns, where invasive species preferentially promote each other over natives (Simberloff & Von Holle, 1999; Zhang, Liu, et al., 2020). One potential mechanism for this stabilizing effect is that, although interaction coefficients and vital rates are not density-dependent, their fluctuations with rainfall could still induce changes in species' endogenous population dynamics (Chesson & Rosenzweig, 1991; Ellner et al., 2019). For example, stabilization

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could occur if species benefiting from rainfall-induced variability in population density exhibit dampened population fluctuations when dominant (Armstrong & McGehee, 1980; Chesson, 1994; Yuan & Chesson, 2015). However, the extent to which this mechanism operates in our system requires further study.

The storage effect fundamentally reflects the interaction between variations in environmental and competitive factors (Chesson, 2000, 2020; Ellner et al., 2019). In recently published studies, λ and α were usually assumed to represent environmental and competitive factors, respectively, with the marginal effects of their joint variation ($\Delta^{\alpha\lambda}$) serving as an analogue to the storage effect (Ellner et al., 2019; Hallett et al., 2019). In our study, $\Delta^{\alpha\lambda}$ showed mixed effects across species pairs, with a marginal net destabilizing effect. However, equating $\Delta^{\alpha\lambda}$ with the storage effect risks misinterpretation. Canonically, the storage effect demands the competitive factor to be density-dependent to generate a systematic rare species advantage: Temporal dominant species face intense competition when conditions favour them, while rare species experience less intense competition during their favourable years (Chesson, 1994; Johnson & Hastings, 2022). In our framework, however, both λ and α depend solely on rainfall scenarios, rendering their covariance fixed and entirely independent of species population densities. As a result, rare species gain no advantage from their rarity.

Although $\Delta^{\alpha\lambda}$ cannot be interpreted directly as the storage effect, we can still ask whether a storage effect might operate in this system. To evaluate that possibility, we examined whether the key preconditions for the storage effect are met (Adler et al., 2006; Chesson & Huntly, 1989). For the storage effect to stabilize coexistence, three requirements must be met: species-specific responses, buffered population growth and covariance between competition and environment (Chesson, 2000). In our system, 6 out of 8 species exhibited statistically similar seed production across rainfall scenarios (Figure S3), violating the first requirement. This response synchrony indicates limited abiotic temporal niche partitioning among these species (Zepeda & Martorell, 2019). Moreover, weak temporal autocorrelation in the rainfall time series (Figure S2) indicates rapid environmental fluctuations. This undermines positive environment-competition covariance (the third requirement), as the environment shifts too quickly—before its past effects on competition (e.g. via population densities) can align with current conditions (Johnson & Hastings, 2022; Schreiber, 2021). Together, these violations suggest that the storage effect may not exert a strong stabilizing influence in our system, though future studies are needed to more fully elucidate its contributions to species coexistence.

Environmental change not only directly affects species performance, but also drives population dynamics indirectly through alterations in interspecific interaction strength (Chu et al., 2016; Van Dyke et al., 2022). The results indicate in our system, competition coefficients varied markedly with rainfall regimes, while intrinsic fecundity remained largely stable (Figure S3). Consequently, environmental variability enhances invasion growth rates primarily through interaction coefficients (Δ^{α}), with vital rates (Δ^{λ}) expected to play a minor role, a prediction that was supported by our findings (Figure 3). This suggests indirect effects of environmental variation outweigh direct effects, aligning with previous research that has quantified both direct and indirect effects of environmental change on population dynamics (Germain et al., 2018; Van Dyke et al., 2022; Wainwright et al., 2019). For instance, Germain et al. (2018) observed that changes in soil moisture did not impact species' intrinsic growth rates but significantly influenced persistence through competition alterations. Findings here underscore the need to incorporate environmental variation when using interaction model to predict species abundances and distributions under climate change.

While previous coexistence studies have predominantly relied on mean or median parameter values to generate binary outcomes (Godoy & Levine, 2014; Kraft et al., 2015; Levine & HilleRisLambers, 2009), our study goes further by accounting for parameter estimation uncertainty to assess coexistence probability (Bowler et al., 2022; Terry & Armitage, 2024). This additional layer of analysis uncovers a continuous spectrum of coexistence potential obscured by point-estimate predictions. For instance, species pairs located outside the coexistence region still maintain non-negligible coexistence probabilities, whereas those within the region exhibit substantial variability: Under variable rainfall, *Ph. angulata*/*Pr. clematidea* shows only a 53.6% likelihood of coexistence, compared to 93.8% for *D. aegyptium*/*L. hyssopifolia*. By capturing nuances missed by binary predictions, our probabilistic approach provides valuable insights for understanding the mechanisms of species coexistence and thus diversity maintenance.

Our controlled experiments provide mechanistic insights into how rainfall variability affects coexistence, but extrapolating these findings to natural systems requires caution due to pot-induced constraints. Restricted rooting volume likely altered species' growth and reproduction strategies. For *O. compositus*, which relies on stolon expansion to exploit resources, the confined space prevented its horizontal spread, likely explaining its competitive disadvantage in our experiments. Meanwhile, *A. conyzoides* and *B. pilosa* exhibited severely suppressed flowering in pots, with many individuals producing no seeds, leading to high uncertainty in their intrinsic growth rate estimates (Figure S3). We also note a limitation associated with extrapolating the Ricker model beyond the density range represented in the experiment. Because the model assumes that interaction coefficients do not vary with density, facilitative effects inferred at low densities may persist unrealistically when resident densities become very high, leading to excessively large invasion growth rates (Buche et al., 2025). However, excluding these extreme outcomes in a conservative sensitivity analysis did not change our main qualitative conclusions (Appendix S5).

While our study focused on rainfall variability, environmental variation encompasses a broader range of abiotic and biotic factors—such as temperature variability and herbivory dynamics—that may jointly influence coexistence (Dee et al., 2020; Jiang & Morin, 2007; Shoemaker et al., 2020). Future work should explore how these concurrent fluctuations interact to shape community dynamics. Moreover, the specific mechanisms mediating the promotion of

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coexistence by rainfall variability—such as functional trait or soil microbial communities—remain unclear (Kandlikar et al., 2022; Wang et al., 2024; Zhang, Hautier, et al., 2020).

In conclusion, our study showed that interannual rainfall variability has a generally positive effect on the coexistence of annual herbaceous plants from fallow subtropical fields. We identified that mechanisms dependent on environmental variability, particularly nonlinearity of interaction coefficients, play a crucial role in promoting coexistence. With the increasing frequency of extreme rainfall and drought events in Southeastern China, where this study took place and other regions of the world (Donat et al., 2016; He & Zhai, 2018; Trancoso et al., 2024), this study offers key insights into the mechanisms that may change the composition of plant communities under climate change scenarios.

AUTHOR CONTRIBUTIONS

Wei Lin and Chengjin Chu conceived and designed the study. Wei Lin conducted the experiments, analysed the data and wrote the first manuscript draft with input from Yuanzhi Li, Margaret M. Mayfield, Lauren Shoemaker, Lauren M. Hallet and Chengjin Chu and finalized the manuscript. Margaret Mayfield also hosted Wei Lin's visit to support the research. All authors interpreted the data and revised the manuscript.

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CONFLICT OF INTEREST STATEMENT

Margaret Mayfield is a Senior Editor, and Yuanzhi Li and Chengjin Chu are Associate Editors of *Journal of Ecology* but took no part in the peer review and decision-making processes for this paper.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/1365-2745.70419.

DATA AVAILABILITY STATEMENT

Data and R-scripts are available in the Dryad Digital Repository at https://doi.org/10.5061/dryad.wh70rxx0f (Lin et al., 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Figure S1. Rainfall pattern at He'er Kou from 1901 to 2020. (a) Average intra-annual rainfall pattern. Pre-flood season spans April–June; post-flood season spans July–September. (b) Historic interannual rainfall records from 1901 to 2020. Years were classified as 'pre-flood dry' (below-average pre-flood, above-average post-flood), 'post-flood dry' (below-average post-flood, above-average pre-flood), 'consistent dry' (below-average in both seasons) or 'consistent wet' (above-average in both seasons).

Figure S2. Partial autocorrelation function (PACF) plot for the annual rainfall time series. The red line represents the PACF estimates, while the blue dashed lines denote the 95% confidence interval. Partial autocorrelations exceeding the blue lines are statistically significant at the 95% level, indicating direct temporal dependence at those lags.

Figure S3. Posterior distributions of intrinsic seed production (λ) and interaction coefficients (α) under four rainfall treatments. The 95% credible intervals for each coefficient are shown as the coloured lines at the bottom of the distributions, with points representing the mean values. '*i*' on the *x*-axis label denotes the focal species. Species code: AGCO, *Ageratum conyzoides*; BIPI, *Bidens pilosa*; DAAE, *Dactyloctenium aegyptium*; LUHY, *Ludwigia hyssopifolia*; OPCO, *Oplismenus compositus*; PHAN, *Physalis angulata*; PRCL, *Praxelis clematidea*; POHY, *Polygonum hydropiper*.

Figure S4. Illustration of invasion growth rate calculation. The x-axis represents the simulation years, and the *y*-axis represents population size. To calculate the invasion growth rate, we simulated the population dynamics of the resident species over a 120-year period under each rainfall scenario. The first 60 years (left of the red dashed line) represent the burn-in phase, allowing the resident's population growth rates to stabilize, with the average growth rate reaching zero. In the following 60 years (right of the red dashed line), we initialized the invader at one in each year, calculated the invasion growth rate, then averaged the invasion growth rates over the 60-year period.

Figure S5. Coexistence probabilities under different rainfall scenarios. Bars represent the mean coexistence probabilities across all species pairs for each rainfall scenario: consistent wet, post-flood dry, pre-flood dry, consistent dry, constant average rainfall and variable interannual rainfall (1901–2020).

Figure S6. Barplot of partitions for each species pair. Median values of the overall invasion growth rate Δ^{r} decomposed into growth rate with no environmental variability (Δ^{0}), main effect of the environmentally driven variation in competition coefficients (Δ^{α}), main effect of the environmentally driven variation in vital rates (Δ^{λ}) and the interaction term between variability in α and λ ($\Delta^{\alpha\lambda}$).

Table S1. Basic information on study species. *g_dry* and *g_wet* represent the seed germination fraction under dry and wet pre-flood season, and *s* denotes the soil bank survival rate.

Table S2. Summary of coexistence probabilities and competitive outcome under constant average and interannual variable rainfall. For each species pair, 1000 parameter sets were drawn from the posterior distributions to estimate invasion growth rates under both rainfall scenarios. Coexistence probability P_{coex} was defined as the proportion of posterior samples in which both species had positive invasion growth rates under two rainfall scenarios. $P_{noncoex_to_coex}$ and $P_{coex_to_noncoex}$ denote the probabilities of a shift in competitive outcome towards or away from stable coexistence, respectively, when moving from constant to variable rainfall. Median outcomes are competitive outcome based on median estimates of invasion growth rate across 1000 posterior samples.

Table S3. Summary of coexistence probabilities and competitive outcome during the mechanism addition analysis. *P*coex represent the probability of coexistence under the variation-independent baseline ($\Delta 0$) and the scenarios where a single variation-dependent mechanism ($\Delta\alpha$ or $\Delta\lambda$) is added. $P_{noncoex_to_coex}$ and $P_{coex_to_noncoex}$

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denote the probabilities of a shift in competitive outcome towards or away from stable coexistence, respectively, when adding a specific mechanism to the baseline. Median outcomes are competitive outcome based on median estimates of invasion growth rate across 1000 posterior samples.

Table S4. Summary of coexistence probabilities and competitive outcome during the mechanism removal analysis. *P*coex represent the probability of coexistence under the full model (Δr) and the scenarios where a single variation-dependent mechanism ($\Delta\alpha$ or $\Delta\lambda$ or $\Delta\alpha\lambda$) is removed. $P_{\text{noncoex_to_coex}}$ and $P_{\text{coex_to_noncoex}}$ denote the probabilities of a shift in competitive outcome towards or away from stable coexistence, respectively, when removing a specific mechanism from the full model. Median outcomes are competitive outcome based on median estimates of invasion growth rate across 1000 posterior samples.

Appendix S2. Model verification.

Figure S7. Posterior predictive check of seed production on the log scale. Red line depicts the density distribution of observed seed production values, while grey lines represent simulated values. Each grey line corresponds to simulated seed production values for one unique draw from the posterior distributions. Because seed counts are inherently non-negative, simulated values below zero produced by the normal error distribution were set to 0 before comparison with the observed data. Subsequently, 0.01 was added to all values prior to log-transformation to avoid undefined logarithms for zero values, resulting in a peak at log(0.01) for observed zero values and simulated values truncated to zero.

Figure S8. Posterior predictive check of seed production on the log scale by species. Red line depicts the density distribution of observed seed production values, while grey lines represent simulated values. Each grey line corresponds to simulated seed production values for one unique draw from the posterior distributions. Because seed counts are inherently non-negative, simulated values below zero produced by the normal error distribution were set to 0 before comparison with the observed data. Subsequently, 0.01 was added to all values prior to log-transformation to avoid undefined logarithms for zero values, resulting in a peak at log(0.01) for observed zero values and simulated values truncated to zero.

Table S5. Model comparison using the Watanabe-Akaike Information Criterion (WAIC). Model forms compared are Ricker (R), Beverton-Holt (BH), Lotka-Volterra (LV) and Law-Watkinson (LW). The best model is determined by the lowest WAIC value and highest model weight, with a substantial advantage indicated when the WAIC difference (dWAIC) exceeds twice its standard error (dSE). The Ricker model is unequivocally the best for all species, as it has the lowest WAIC and the full model weight.

Appendix S3. Key results from the model with pot as a random effect.

Figure S9. Comparisons of parameter estimates between models with and without pot as a random effect. The 95% credible intervals for each coefficient (i = focal species) are shown as the transparent coloured lines, with points representing the mean values. The *x*-axis represents the estimated coefficients from models without the random effect, while the *y*-axis represents the estimated coefficients from models including pot as a random effect.

Figure S10. Pairwise coexistence outcomes under constant and variable rainfall scenarios, based on a model with pot included as a random effect. Points denote median invasion growth rates of species 1 and 2 (first and second in pair labels) when they invade each other's equilibrium population, calculated from 1000 parameter combinations drawn from posterior distributions. Pie charts overlaid on each point depict the posterior probabilities of each coexistence outcome (coexistence, priority effect, species 1 wins and species 2 wins). These values are displayed under constant rainfall scenarios (consistent wet, post-flood dry, pre-flood dry, consistent dry, average) and variable interannual rainfall scenario for each species pair. Pairs coexist when both invasion growth rates exceed 0, as indicated by the light green region. Pairs with invasion growth rates exceeding the displayed range are marked with arrows.

Figure S11. Coexistence probabilities under different rainfall scenarios, based on a model with pot included as a random effect. Bars represent the mean coexistence probabilities across all species pairs for each rainfall scenario: consistent wet, post-flood dry, pre-flood dry, consistent dry, constant average rainfall and variable interannual rainfall (1901–2020).

Figure S12. Role of different variation-dependent coexistence mechanisms, based on a model with pot included as a random effect. Points denote median Δ partition terms representing three variation-dependent mechanisms for species 1 and 2. The variation-dependent mechanisms include main effect of environmentally driven variation in interaction coefficients (Δ^{α}, relative nonlinearity in interaction coefficients), main effect of environmentally driven variation in vital rates (Δ^{λ}, relative nonlinearity in intrinsic growth rate) and interaction effect between variation in interaction coefficients and vital rates ($\Delta^{\alpha\lambda}$).

Figure S13. The contribution of different variation-dependent coexistence mechanisms to coexistence probability, based on a model with pot included as a random effect. Grey points show the coexistence probabilities for each species pair as three variation-dependent mechanisms (Δ^{α}, Δ^{λ}, $\Delta^{\alpha\lambda}$) are added stepwise from Δ^{0} to Δ^{r}. Bold green points and line represent the average coexistence probability across 28 species pairs.

Appendix S4. Key results from analysis incorporating uncertainty in soil seed bank survival (*s*) and germination fraction (*g*).

Figure S14. Pairwise coexistence outcomes under constant and variable rainfall scenarios, incorporating uncertainty in soil seed bank survival (*s*) and germination fraction (*g*). Points denote median invasion growth rates of species 1 and 2 (first and second in pair labels) when they invade each other's equilibrium population. Invasion growth rates were calculated using 1000 parameter sets, with interaction coefficients (α) and intrinsic seed production rates (λ) sampled from posterior distributions, and soil seed bank survival (*s*) and germination fractions (*g*) drawn from binomial distributions. Pie charts overlaid on each point depict the posterior probabilities of each coexistence outcome (coexistence, priority effect, species 1 wins and species 2 wins). These values are displayed under constant

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rainfall scenarios (consistent wet, post-flood dry, pre-flood dry, consistent dry, average) and variable interannual rainfall scenario for each species pair. Pairs coexist when both invasion growth rates exceed 0, as indicated by the light green region. Pairs with invasion growth rates exceeding the displayed range are marked with arrows.

Figure S15. Coexistence probabilities under different rainfall scenarios, incorporating uncertainty in soil seed bank survival (*s*) and germination fraction (*g*). Bars represent the mean coexistence probabilities across all species pairs for each rainfall scenario: consistent wet, post-flood dry, pre-flood dry, consistent dry, constant average rainfall and variable interannual rainfall (1901–2020).

Figure S16. Role of different variation-dependent coexistence mechanisms, incorporating uncertainty in soil seed bank survival (*s*) and germination fraction (*g*). Points denote median Δ partition terms representing three variation-dependent mechanisms for species 1 and 2. The variation-dependent mechanisms include main effect of environmentally driven variation in interaction coefficients (Δ^{α}, relative nonlinearity in interaction coefficients), main effect of environmentally driven variation in vital rates (Δ^{λ}, relative nonlinearity in intrinsic growth rate) and interaction effect between variation in interaction coefficients and vital rates ($\Delta^{\alpha\lambda}$).

Figure S17. The contribution of different variation-dependent coexistence mechanisms to coexistence probability, incorporating uncertainty in soil seed bank survival (*s*) and germination fraction (*g*). Grey points show the coexistence probabilities for each species pair as three variation-dependent mechanisms (Δ^{α}, Δ^{λ}, $\Delta^{\alpha\lambda}$) are added stepwise from Δ^{0} to Δ^{r}. Bold green points and line represent the average coexistence probability across 28 species pairs.

Appendix S5: Key results from the analysis excluding pairs with mean invasion growth rates >10,000.

Figure S18. Coexistence probabilities under different rainfall scenarios, based on the subset of species pairs with mean invasion growth rates below 10,000. Bars represent the mean coexistence probabilities across all species pairs for each rainfall scenario: consistent wet, post-flood dry, pre-flood dry, consistent dry, constant average rainfall and variable interannual rainfall (1901–2020).

Figure S19. Role of different variation-dependent coexistence mechanisms, based on the subset of species pairs with mean invasion growth rates below 10,000. Points denote median Δ partition terms representing three variation-dependent mechanisms for species 1 and 2. The variation-dependent mechanisms include main effect of environmentally driven variation in interaction coefficients (Δ^{α}, relative nonlinearity in interaction coefficients), main effect of environmentally driven variation in vital rates (Δ^{λ}, relative nonlinearity in intrinsic growth rate) and interaction effect between variation in interaction coefficients and vital rates ($\Delta^{\alpha\lambda}$).

Figure S20. The contribution of different variation-dependent coexistence mechanisms to coexistence probability, based on the subset of species pairs with mean invasion growth rates below 10,000. Grey points show the coexistence probabilities for each species pair as three variation-dependent mechanisms (Δ^{α}, Δ^{λ}, $\Delta^{\alpha\lambda}$) are added stepwise from Δ^{0} to Δ^{r}. Bold green points and line represent the average coexistence probability across 28 species pairs.

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